

Compare SILcet with exSILentia

To perform the SIL verification in process industry, according to IEC 61508 and IEC 61511, there are several software tools such as “exSILentia” by Exida, our SILcet software and others.

In the process industry “exSILentia” is, without a doubt, the **reference software with great international recognition**. It is a great suite of applications, including “SILver” (SIL verification), which is the tool we want to compare with SILcet.

Each tool has its advantages and disadvantages. The user must analyze in each case **which one best suits their needs**. The decision should not be based on the results of the calculations but on other criteria, because, as we will see in the examples at the end, the calculated values of PFDavg, RRF and MTTFs are very similar with either of the two tools.

Let us briefly see what, in our opinion, are the **main differences between SILcet and exSILentia**.

Advantages of exSILentia online

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Excel version of SILcet needs **Windows and Excel**; to use it on a Mac you have to do so in a Virtual Machine; the SILcet Cloud version only needs a **browser and an internet connection**).

- Integrates **other applications** (SRS, LOPA, SILect).
- **Integrated database of more than 200 manufacturers** with the data used in the calculations (failure rates, etc.). SILcet allows the user to create their own database with the SIF elements used in the project. SILcet Cloud also includes an extensive database.
- Generates main **reports in Word format** (SILcet generates reports in PDF and Excel).
- The **user interface** is better in the web version than in Excel, due to Excel's limitations in this regard.

(1)

> Probability (1)

> Sensor (2)

> SIF (11)

> SILcet (2)

Advantages of SILcet

- There are **2 versions**, one **based on Excel** (SILcet Pro Plus), and another **web-based** (SILcet Cloud). **Both use the same calculation algorithms**, meaning they generate the same results.
- The **license price of SILcet is much lower**.
- If there is **no good internet connection**, the **Excel-based version is a good alternative** (the

They can be generated in PDF format or in Excel.

Furthermore, **reports can be generated in any language.**

- SILcet allows to select a **different Route (1H/2H/IEC61511) for each element of the SIF.**
- In SILcet Cloud the complete project with several SIFs can be **exported to Excel, which allows it to be modified in Excel and imported back.**
- SILcet Cloud allows **exporting to Excel, and importing, the databases** created by the user.
- SILcet allows **selecting a different Route (1H/2H/IEC61511) for each element of the SIF.**
- When the **license expires**, SILcet Pro Plus, based on Excel, can be opened and its content viewed.
In SILcet Cloud, all content can be seen in the exported Excel file

In the following table we show **calculation results** of several examples **performed with both tools.** We observe that **they are very similar.**

In both cases it is necessary that the user has knowledge of Functional Safety in order to enter the data correctly. **Both tools are very mature and have been well tested.**

final element	1002	2,18E-03	75	2,46E-03	73	TI= 1y. / MTTR= 24h. Final element-2: relay + contactor PTC (Cpt)=95% / Beta= 10% TI= 1y. / MTTR= 24h.
logic solver	1002D	1,67E-04	171	1,39E-04	192	
SIF-2	PFDavg RRF	2,37E-03 421,5	51,4	2,60E-03 384,8	65	Logic Solver: safety PLC PTC (Cpt)=99% / Beta= 2% TI= 4y. / MTTR= 24h. (To compare properly, the same logic solver has been used in all cases)
sensor	1001	3,36E-04	334	3,62E-04	334	
final element	2002 (1002 & 1001)	6,69E-03	2553	6,99E-03	127	
logic solver	1002D	1,67E-04	171	1,39E-04	192	
SIF-3	PFDavg RRF	7,19E-03 139,0	108,3	7,49E-03 133,5	62	
sensor	1002	1,69E-05	171,17	1,83E-05	171	
final element	1002 (1002 OR 1001)	9,82E-06	44,35	1,30E-05	41	
logic solver	1002D	1,67E-04	171	1,39E-04	192	
SIF-4	PFDavg RRF	1,94E-04 5162,1	29,2	1,70E-04 5872,0	28	

To know more you can [download this technical paper](#).

Information regarding SILcet Certification

- **Certification Status:** SILcet is not yet certified by a third-party body. We are currently working toward certifying the upcoming **SILcet Cloud** version soon.
- **Standards Compliance:** The IEC 61508 and IEC 61511 standards **do not require** the use of certified software tools for performing Safety Integrity Level (SIL) verification calculations. It is acceptable, according to the standards, to even perform these calculations using generic tools like Microsoft Excel.
- **Maturity and Reliability:** SILcet is a highly mature and robust product that has been commercially available since 2018. The software is developed based on the rigorous requirements of recognized international standards:

- IEC 61511 (Functional safety – Safety instrumented systems for the process industry sector)
- ISA TR84.00.02 (Safety Integrity Level (SIL) Verification of SIS)
- **A Note about other tools:** The current version (v4.X) of exSILentia is not certified as a software product by an independent third party. The credibility of exSILentia primarily derives from its developer, exida, which is an accredited certification body recognized by **ANSI** (American National Standards Institute). exSILentia is a mature and widely used tool, with pros and cons like any other product.

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